

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

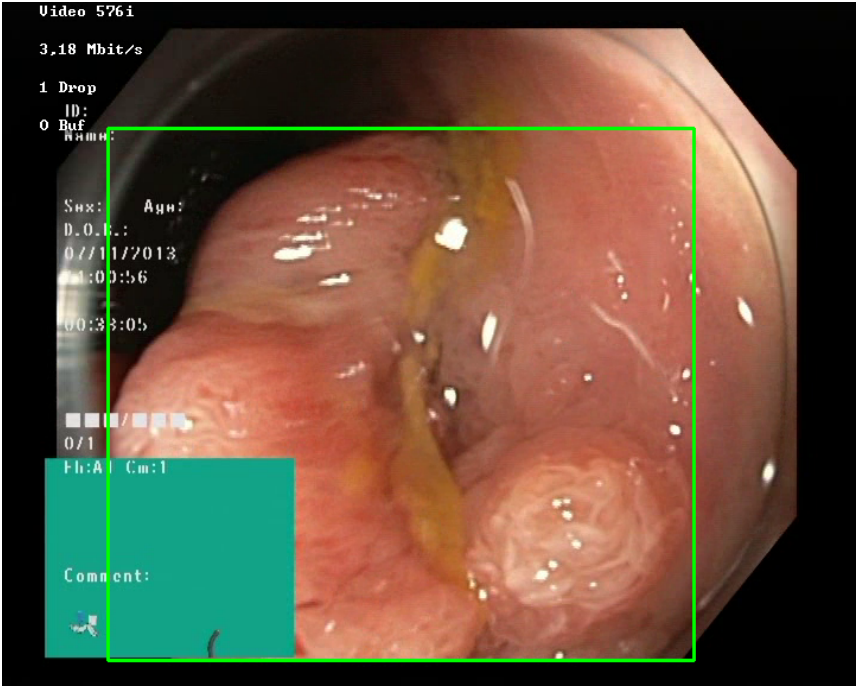
Video ID:	3a7f1bb3-9e72-468a-be4a-35bdbc3a2c5d
Total Frames:	3006
Frame Rate:	25.0 fps (assumed)
Duration:	120.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1591	✓
RT-DETR Detections	1949	✓
MedSAM2 Detections	1968	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	3	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 828 frames)



Feature	Value	Interpretation
Frame Index	493	Frame 493 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_P OLYP	Type confidence: 92.0%
Redness Score	0.164	Color-based vascularization
Relative Radius	58.9%	Polyp size as % of frame diagonal
Texture Score	0.279	Surface morphology
Vessel Visibility	0.676	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output

Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 21 frames)

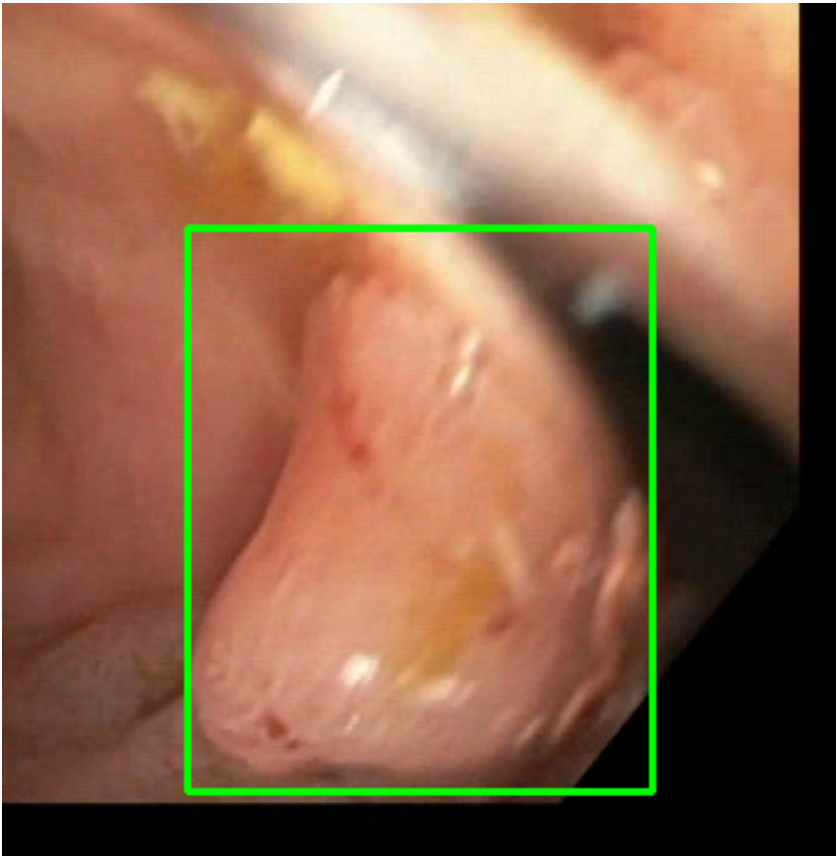


Feature	Value	Interpretation
Frame Index	2254	Frame 2254 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.075	Color-based vascularization
Relative Radius	13.3%	Polyp size as % of frame diagonal
Texture Score	0.375	Surface morphology
Vessel Visibility	0.618	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.1%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 89 frames)



Feature	Value	Interpretation
Frame Index	2972	Frame 2972 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 89.1%
Redness Score	0.138	Color-based vascularization
Relative Radius	38.1%	Polyp size as % of frame diagonal
Texture Score	0.216	Surface morphology
Vessel Visibility	0.350	Vascularization indicator

Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	77.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 3 (100.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.